

1 Introduction

Viterbo’s conjecture [Vit00] states that among convex bodies of a given volume in $\mathbb{R}^{2n}$ the Euclidean ball has the largest symplectic capacity. We use the Ekeland–Hofer–Zehnder (EHZ) capacity. For a convex body K with smooth boundary, $c_{\text{EHZ}}(K)$ is the least action $\int_\gamma \lambda$ among closed characteristics γ on ∂K , that is, closed curves tangent to the line field $\ker(\omega_0|_{T\partial K})$; here $\omega_0 = \sum_j dq_j \wedge dp_j$ and $d\lambda = \omega_0$. For nonsmooth convex bodies, polytopes in particular, the capacity is the limit over smooth approximations. In dimension four the conjecture reads $\text{sys}(K) \leq 1$ for the *systolic ratio*

$$\text{sys}(K) = \frac{c_{\text{EHZ}}(K)^2}{2 \text{vol}_4(K)},$$

which equals one on balls and is invariant under translations, rescalings and linear symplectic maps.

Haim–Kislev and Ostrover disproved the conjecture with a Lagrangian product [HO26]. Write $\mathbb{R}^4 = \mathbb{R}_q^2 \oplus \mathbb{R}_p^2$. For planar convex bodies K_q, K_p , the Lagrangian product $K_q \times_L K_p$ is the set of (q, p) with $q \in K_q$ and $p \in K_p$; each factor lies in a Lagrangian plane. Let P_5 be the regular pentagon of circumradius one centered at the origin with a vertex at $(1, 0)$, and $R(\theta)$ the rotation by θ . The body

$$K_{\text{HKO}} = P_5 \times_L R(-\pi/2)P_5 \quad \text{has} \quad \text{sys}(K_{\text{HKO}}) = \frac{3 + \sqrt{5}}{5} \approx 1.047.$$

This thesis starts from the counterexample and asks how much progress a fixed toolbox makes on the questions around it. The toolbox consists of high-performance implementations of capacity algorithms, proofs by exact computation, standard data-science methods, computing time on the LICCA cluster of the University of Augsburg, and AI agents. The questions came out of the work. For example, random perturbations of K_{HKO} did not increase its systolic ratio. This suggested that K_{HKO} is a local maximum, and for perturbations that keep ten facets this is now the main theorem of the thesis. The results therefore answer different questions and are presented side by side rather than as one argument.

One of the earliest questions was whether standard data science finds new, interesting bodies with systolic ratio above one. It did not. Bodies above one arise near K_{HKO} , since the systolic ratio is continuous, but random sampling, adaptive sampling and local search started from random bodies all stayed below one. The data did show a pattern: bodies whose two-dimensional faces have small total symplectic area, relative to volume, tend to have large systolic ratio. Looking for patterns in data has helped pure mathematics before; in knot theory, relations noticed in tables of computed invariants were later proved. Here the approach goes one step further: AI agents ran the data science and also did large parts of the theory work. For instance, from the suggestion to control a neighborhood of K_{HKO} by several upper bounds on the systolic ratio, a GPT-5-series model found a configuration of twenty-six such bounds that suffices, and wrote the exact SageMath verification. AI agents are therefore part of the method, and the thesis reports what they contributed.

The main results are the following. Every ten-facet polytope sufficiently close to K_{HKO} has systolic ratio at most $(3 + \sqrt{5})/5$, with equality only for images of K_{HKO} under translations, rescalings and linear symplectic maps. The ten-facet space has another local maximum: a rational product of a hexagon and a quadrilateral with systolic ratio one, which is therefore not a new counterexample. No product of two independently linearly deformed regular pentagons exceeds the systolic ratio of K_{HKO} . Symplectic ridge area has a strong inverse rank association with the systolic ratio in sampled populations, but exact formulas for rotated pentagon and triangle products order the two quantities in opposite ways. Behind these results stand Haim–Kislev’s finite capacity formula [Hai19], its first-order variation, and a second capacity algorithm built on the local passage geometry of Chaidez and Hutchings [CH21].

The rest of this introduction first places K_{HKO} among capacities, dynamics and convex geometry, then describes the results in more detail and ends with a guide to the chapters.

References

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